

Unit

2

Lesson

3



Innovators in Action.

Alert Mode On!

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Compare ultrasonic and infrared sensors based on real testing.
- Decide which sensor is better for a smart walker
- Build and program a working obstacle-detection system using Arduino.
- Attach a vibration motor that alerts the user of danger and Bluetooth module to send message to caregiver.
- Compare ultrasonic and infrared sensors based on real testing.

Challenge questions of the lesson

- How do ultrasonic sensors and infrared sensors compare in terms of accuracy and effectiveness for detecting obstacles in a smart walker?
- Which sensor—ultrasonic or infrared—provides the most reliable performance for an obstacle-detection system in a smart walker?
- How can you design and program an effective obstacle-detection system using Arduino?
- What are the benefits of attaching a vibration motor to the system, and how does it help alert the user of potential dangers?
- How can integrating a Bluetooth module into the system improve the functionality of the smart walker by sending alerts to a caregiver?

SEL Relation

- **Self-Awareness:** Students reflect on their decisions when comparing sensors, developing critical thinking skills.
- **Self-Management:** Building and programming the system helps students manage time, set goals, and persist through challenges.
- **Social Awareness:** The design process fosters empathy by considering how technology can improve the lives of users, especially the elderly.
- **Relationship Skills:** Collaborating on testing and designing promotes teamwork and communication.
- **Responsible Decision-Making:** Students make ethical choices about safety and the impact of their design on users.

Engage



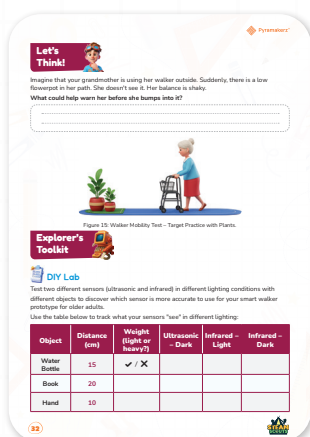
Teacher's Role:

- Facilitator of curiosity and empathy.
- Guide for student thinking without giving answers.
- Model for inquiry, emotional connection, and inclusive participation.
- Encourager of all learners to share in different ways.

Expected From Students:

- Use empathy to consider real-world challenges.
- Ask "What if..." and "How might we..." questions.
- Think about potential technologies that could help.
- Begin to form hypotheses about tools/devices (without needing prior technical knowledge).
- Engage through discussion, drawing, storytelling, or quiet reflection.

Parts of the book included.



Let's Think!



Objective:

Students will develop empathy and creative problem-solving by imagining assistive features for a walker that could help an elderly person avoid obstacles, especially when vision or hearing is limited.

Possible Students Answers:

- "Maybe the walker could beep when something is close in front of it."
- "It could have a sensor that vibrates when an object is nearby."

- "What if there was a tiny camera that sees what's on the ground and warns her?"
- "The walker could light up or flash when it detects something in the way."
- "She might not hear well, so maybe it can shake a little to get her attention."
- "What if it had a voice that said, 'Object ahead'?"
- "A sensor could tell how far something is, and warn her if it's really close."
- "The walker could send a signal to her phone or a smartwatch to alert her."
- "It should work even if it's dark outside or if the flowerpot is small."
- "It could have wheels that stop moving if something is blocking the path."

Activity Instructions:

Let students:

- Think quietly for 1–2 minutes
- Sketch or write one idea on a sticky note
- Share ideas in pairs or groups
- Discuss: How would she know something is in her way without seeing it or hearing it?

Possible Strategies for Implementation:

- **Class Discussion:**
 - Begin with a short conversation about challenges older adults face when walking.
 - Ask students to share ideas on how technology can help prevent accidents.
- **Think-Pair-Share:**
 - Students first think of their own solutions, then discuss with a partner, and finally share with the class.
- **Visual Brainstorming:**
 - Provide chart paper or a whiteboard where students can draw or list possible warning systems (e.g., beeping sensors, flashing lights).
- **Technology Connection:**
 - Show examples of ultrasonic or infrared sensors and explain how they could be added to a walker.
- **Prototype Sketch:**
 - Have students create a quick sketch of a walker with a built-in alert system.
- **Link to Real-World Devices:**
 - Display images or videos of assistive technologies already used for obstacle detection.

Explorer's Toolkit



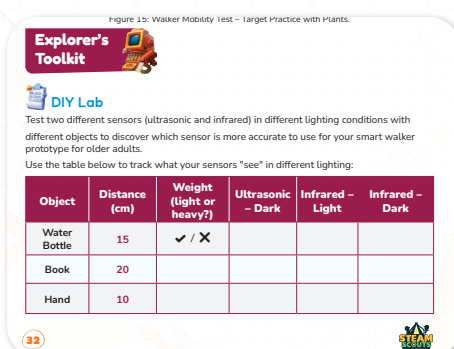
Teacher's Role:

- Facilitator of exploration – encourages curiosity rather than giving direct answers.
- Team coach – supports collaboration, assigns/challenges roles, and promotes inclusive participation.
- Thinking partner – asks open-ended questions to deepen understanding.
- Resource guide – provides circuit diagrams, tools, or virtual simulations as needed.
- Emotional coach – supports students through frustration and celebrates effort and teamwork.
- Key teacher actions: Monitor progress, clarify confusion without solving problems directly.
- Ask, “What do you notice when you switch the sensor?” or “How does the light affect your results?”
- Offer scaffolds: visual diagrams, a working demo, or peer-led help stations.
- Remind students that “wrong” setups are part of real engineering work — valuable learning moments.
- Encourage diverse talents: Some students may lead circuit work; others may organize data or narrate findings.
- Guide teams to collaborate by rotating roles or sharing results.

Expected From Students:

- Work in teams to build circuits using both ultrasonic and infrared sensors.
- Test each sensor in two different lighting conditions: normal and dim/covered.
- Record observations using a results table (symbols, numbers, sketches, or brief notes).
- Compare sensor performance based on range, accuracy, and consistency.
- Collaborate respectfully in roles: builder, tester, observer, and recorder.
- Reflect on: Which sensor would keep someone safer on a smart walker — and why?

Parts of the book included.



Objective:

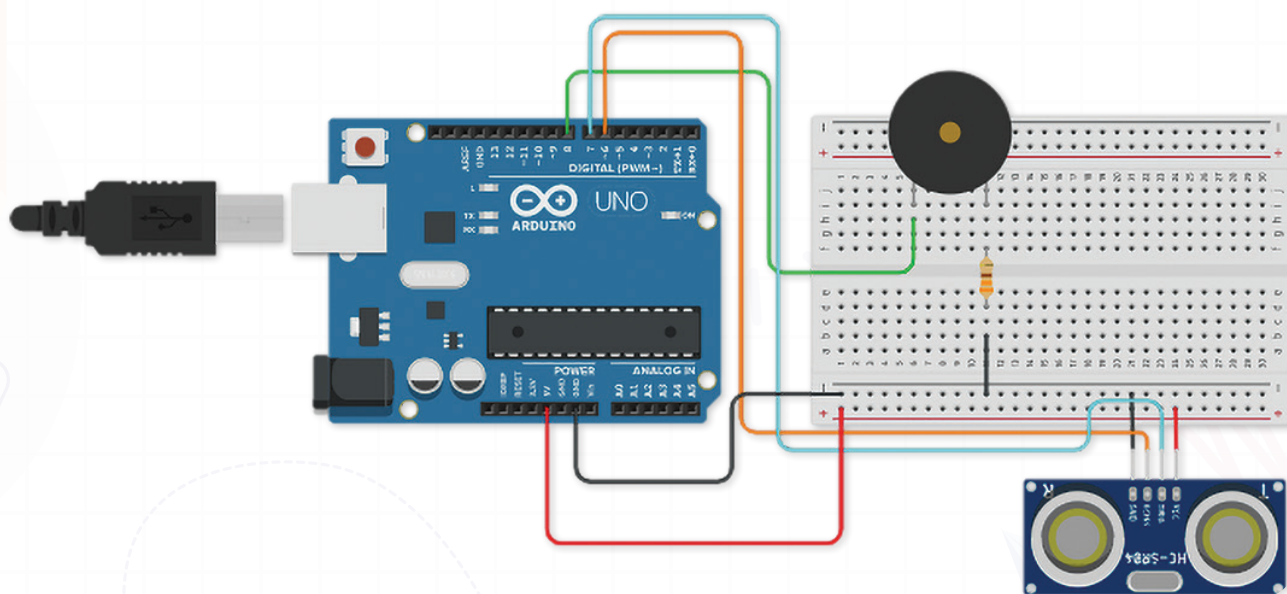
Test two types of sensors (ultrasonic and infrared) in light and dark to find out which one works better for helping an elderly person avoid obstacles with a Smart Walker.

Materials Needed:

- 1 Arduino Uno board (pretested)
- 1 ultrasonic sensor (e.g., HC-SR04)
- 1 infrared sensor (e.g., IR obstacle avoidance module)
- Jumper wires
- Breadboard
- Measuring tape or ruler
- Objects to test (book, hand, water bottle)
- Light source (flashlight or lamp for simulation)
- Dim area in the classroom (or box to block light)
- Preloaded Arduino code

Activity Instructions:**1. Set Up Your Sensor Test Station**

- Connect the ultrasonic sensor to your Arduino and breadboard.



- Check your wiring and upload your code.

```
// Define pins
const int trigPin = 9;
const int echoPin = 10;
const int buzzerPin = 8;
// Variables for duration and distance
long duration;
int distance;
void setup() {
  // Set pin modes
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(buzzerPin, OUTPUT);
  // Start serial monitor
  Serial.begin(9600);
}
void loop() {
  // Clear the trigPin
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  // Send a 10us pulse to trigPin
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  // Read the echoPin, return sound wave travel time in microseconds
  duration = pulseIn(echoPin, HIGH);
  // Calculate distance in cm
  distance = duration * 0.034 / 2;
  // Print distance to Serial Monitor
  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");
  // If object is closer than 20 cm, sound the buzzer
  if (distance < 20) {
    tone(buzzerPin, 1000); // Play tone at 1kHz
```

```

} else {
  noTone(buzzerPin); // Stop the tone
}

delay(200); // Small delay for stability
}

```

2. Test in Normal Room Light

- Place your test object (book, hand, bottle) 10–30 cm from the sensor.
- Slowly move the object closer or farther.
- Watch how the sensor responds.
- Record the distance when the object is first detected.

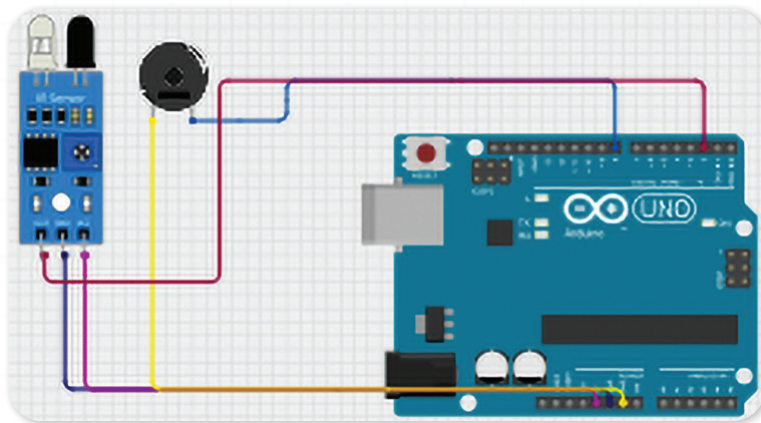
3. Repeat in Low Light

- Turn off lights or move to a dim corner.
- Test the same object again with the same sensor.
- Record what changes in detection (if any).

4. Switch to Infrared Sensor

- Replace the sensor and repeat Steps 2 & 3 using the infrared sensor.

Circuit:



Code:

```

// Define pins
const int irSensorPin = 2; // IR sensor output pin
const int buzzerPin = 8; // Buzzer pin
int sensorValue = 0; // Variable to store IR sensor value
void setup() {
  pinMode(irSensorPin, INPUT); // Set IR sensor pin as input
  pinMode(buzzerPin, OUTPUT); // Set buzzer pin as output
  Serial.begin(9600); // Start serial communication
}

```

```

}
void loop() {
  // Read the sensor value
  sensorValue = digitalRead(irSensorPin);
  // Print value to Serial Monitor
  Serial.print("IR Sensor: ");
  Serial.println(sensorValue);
  // If IR sensor detects an obstacle (LOW = object detected)
  if (sensorValue == LOW) {
    tone(buzzerPin, 1000); // Turn on buzzer with 1kHz tone
  } else {
    noTone(buzzerPin); // Turn off buzzer
  }
  delay(100); // Small delay for stability
}

```

Possible Answers:

Object	Distance (cm)	Ultrasonic – Light	Ultrasonic – Dark	Infrared – Light	Infrared – Dark
Water Bottle	15	✓	✓	☑	✗
Book	20	✓	✓	☑	✗
Hand	10	✓	✓	☑	✗

Reflection Responses:

Note to Teacher: No right or wrong answers Students are free to choose ultrasonic sensors or infrared sensors in their smart walker prototypes

- **Claim:** I believe the ultrasonic sensor is more accurate and reliable for use in a Smart Walker.
- **Evidence:** When we tested with the water bottle, book, and hand, the ultrasonic sensor consistently detected obstacles, even in both light and dark conditions.
- **Reasoning:** The data shows that the ultrasonic sensor worked better because it was not affected by changes in lighting. This makes it a safer and more dependable choice for elderly people, especially if they are using the walker in dimly lit or outdoor nighttime environments.

Possible Strategies for Implementation:

1. Hands-On Sensor Stations

- Set up two workstations: one with an ultrasonic sensor circuit, one with an infrared sensor circuit.
- Students rotate between stations, testing both sensors with different objects and lighting conditions.

2. Simulated Real-World Scenarios

- Dim the lights or use a shaded area to simulate “dark” conditions.
- Place obstacles at different distances to test detection accuracy.

3. Group Data Collection & Sharing

- Students work in small teams to record results in the provided table.
- After testing, teams share results on a class board or digital spreadsheet for comparison.

4. Guided Data Analysis

- Teacher leads a short discussion on patterns observed (e.g., sensor sensitivity, effect of light).
- Students highlight any anomalies and discuss possible reasons.

5. CER Modeling

- Teacher models how to write a claim, evidence, and reasoning using one example object.
- Students complete their own CER based on their group’s data.

6. Peer Review

- Students swap CER responses with another group and give feedback based on clarity and use of evidence.

7. Extension Challenge

- Ask students to brainstorm how both sensors could be combined in the walker for even better safety.

Explain



Teacher’s Role:

- Introduce the purpose of the video and article, explaining how they relate to the Smart Walker project.
- Guide students to focus on key points:
 1. In the video: what a vibration motor does, how it works, and possible uses for assisting elderly users.

2. In the article: the functional differences between infrared and ultrasonic sensors, including strengths and weaknesses.

- Pause the video at important moments to ask comprehension questions.
- Encourage students to take short notes or highlight important terms while reading the article.
- Facilitate a brief discussion after both resources to ensure understanding.

Expected From Students:

- Watch the vibration motor video attentively and identify at least two possible uses for it in the Smart Walker.
- Read the article and summarize the main differences between IR and ultrasonic sensors.
- Note down one advantage and one limitation for each sensor type.
- Share findings with a partner or small group and discuss which sensor might be better for elderly safety in different environments.
- Ask clarifying questions if they find a term or concept confusing.

Parts of the book included.



Screen Lessons
 Let's watch this video to learn more about vibration motor, it's function and its connections.
<https://www.youtube.com/watch?v=INVYx8f2RIs>





Screen Lessons

Objective:

- Learn how a vibration motor works and connects to Arduino.
- Understand how sensors help detect obstacles.
- Connect video ideas to the Smart Walker project.

Video Link: <https://www.youtube.com/watch?v=INVYx8f2RIs>



Possible Strategies for Implementation:

1. Pre-Viewing

- **Activate Prior Knowledge:** Ask students if they've ever felt their phone vibrate or used a device with a vibration alert. Discuss why vibrations might be useful for people with hearing or vision challenges.
- **Set a Purpose for Watching:** Tell students to watch for three things: what the vibration motor does, how it works, and how it can be connected to a circuit.
- **Prediction Question:** "How do you think a vibration motor could help make the Smart Walker safer for elderly people?"

2. During Viewing

- **Focused Viewing:**
 - Pause at key points to clarify terms like “motor function,” “power source,” or “polarity.”
 - Ask quick check-in questions: “What part of the circuit sends power to the motor?”
- **Visual Notes:** Have students sketch a quick diagram of the vibration motor and label its parts while watching.
- **Highlight Applications:** Point out moments when the video shows real-world uses, linking them to the Smart Walker design.

3. After Viewing

- **Think-Pair-Share:** Students discuss in pairs how the vibration motor could be integrated into the walker to give warnings.
- **Hands-On Connection:** If materials are available, let students see or handle a vibration motor before building their own circuits in later lessons.
- **Reflection Prompt:** “If you were designing a walker, when would you make the vibration motor activate?”
- **Link to Project:** Record ideas in the project design journal for use in the Smart Walker prototype phase.



Dig Deeper

Objective:

- Learn how ultrasonic and infrared sensors work.
- Compare which sensor works better in different conditions.
- Choose the best sensor for the Smart Walker and explain why.

Article link: https://maxbotix.com/blogs/blog/ultrasonic-or-infrared-sensors?srsId=AfmBOpaLLsckBQpdSumY8zQfpxcumYPJqOz_td508D1tFdbDUNdQiqc



Possible Strategies for implementation:

1. Pre-Reading Activation

- Ask students: “What differences do you already know between infrared and ultrasonic sensors?”
- Show images of both sensors and have students make quick predictions about which might work better in different environments.

2. Guided Reading with Focus Questions

- Provide students with 3–4 guiding questions to answer as they read, such as:
 - What is one advantage of infrared sensors?
 - What is one advantage of ultrasonic sensors?
 - In what situations might each sensor fail?

- Encourage them to highlight or take notes while reading.

3. Jigsaw Approach

- Divide the article into sections (e.g., “Infrared Basics,” “Ultrasonic Basics,” “Comparison”) and assign each group a section to read.
- Groups share their findings with the class to complete a comparison chart together.

4. Comparison Table

- Have students create a side-by-side chart of features, pros, and cons of each sensor.
- Use the table later to help justify their sensor choice in the Smart Walker design.

5. Real-World Connection

- After reading, ask: “If you were designing the Smart Walker for use both indoors and outdoors, which sensor would you choose and why?”
- Let students defend their answers in small group discussions.

6. Link to Testing Activity

- Remind students they will soon test both sensors. Ask them to predict if the article’s claims will match their hands-on results.

Elaborate



Teacher’s Role:

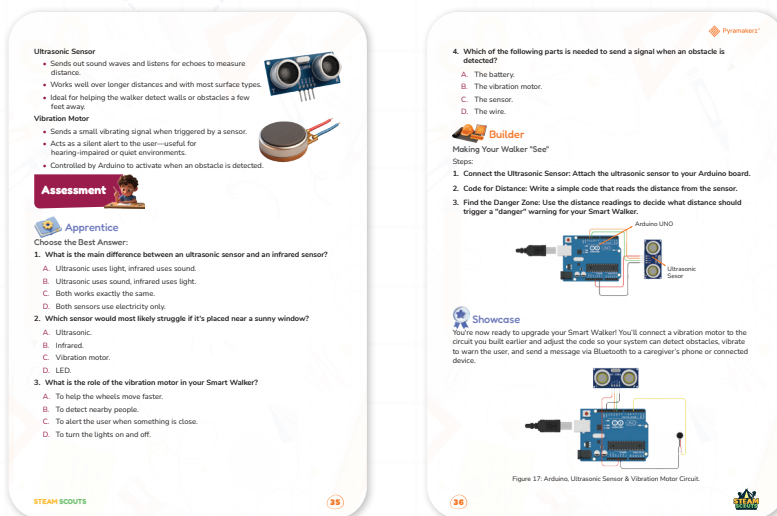
- Facilitate student engagement with the MCQs as a formative check, not a test.
- Prompt critical thinking by inviting student reasoning behind choices.
- Support students in working independently, in pairs, or groups — offering different ways to access the content.
- Celebrate process over perfection, reinforcing the mindset that engineers test and learn.
- **Guide and Mentor:** Provide support and answer questions as students work on the circuit and code.
- **Trouble-shooter:** Help students diagnose issues with wiring or code without giving direct answers.
- **Safety Monitor:** Ensure safe handling of electronic components and power.
- Encourage Creativity and Application of Knowledge: Prompt students to think about how they can adapt or extend the basic setup for real-world scenarios.

Expected From Students:

- Read and answer each MCQ thoughtfully.
- Reflect on their confidence in their choices.
- Participate in discussions that explore why an answer is correct.

- Use their results to inform their design decisions in the upcoming “Practice” and “Showcase” parts.
- **Apply Technical Skills:** Successfully create circuits and write code for microcontrollers.
- **Problem-Solve:** Debug wiring connections and code errors independently or collaboratively.
- **Experiment and Iterate:** Test their setup, identify danger zones, and refine their code for accuracy.
- **Create Models, Circuits, Experiments, or Code:** Build the physical circuit, write the Arduino program, and conduct tests to make the walker “see.”

Parts of the book included.



Apprentice

Objective:

Students will demonstrate their understanding of the differences between ultrasonic and infrared sensors, identify potential limitations of each, and explain the role of a vibration motor in the Smart Walker system. This will reinforce their knowledge of how components interact to detect and alert users to obstacles.

Answers:

- A. Ultrasonic uses sound, infrared uses light
- B. Infrared
- C. To alert the user when something is close
- D. The sensor

Possible Strategies for Implementation:

- **Review Before Quiz**
 - Have a quick recap discussion of ultrasonic vs. infrared sensors and the vibration motor’s role before students attempt the questions.
- **Peer Quiz Practice**
 - Students quiz each other in pairs before answering independently.

- **Interactive Clickers or Polling**
 - Use Kahoot, Mentimeter, or simple hand-raising to make the quiz more engaging.
- **Think-Aloud**
 - Model answering one question by verbalizing your thought process (“I know infrared uses light because...”).
- **Feedback Loop**
 - Go through the answers as a class, asking students to explain why an answer is correct instead of just giving the letter.
- **Extension**
 - After completing the quiz, have students write one new multiple-choice question related to sensors or vibration motors for their peers to answer.



Builder

Objective:

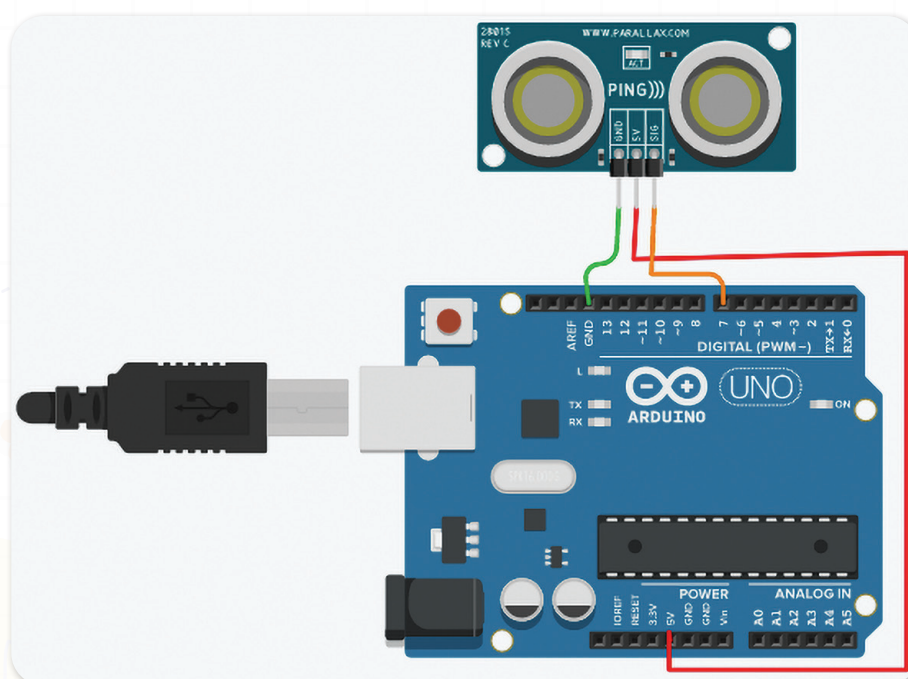
Students will integrate and program an ultrasonic sensor with Arduino to measure distances and define a “danger zone” for obstacle detection in the Smart Walker prototype.

Answer / Expected Outcome:

- Students successfully connect the ultrasonic sensor to the Arduino board.
- They write and upload a code that displays distance readings in the Serial Monitor.
- They determine a distance threshold (e.g., 20 cm) that will act as a warning zone for the Smart Walker.

Activity Instructions:

1. **Connect the Ultrasonic Sensor:** Attach the sensor to the Arduino following the provided wiring diagram.



2. Code for Distance: Write a program that:

- Sends a pulse from the ultrasonic sensor.
- Reads the time taken for the echo to return.
- Calculates and displays the distance in centimeters.

```
// Define the pin for the PING sensor
const int pingPin = 7; // Signal pin connected to digital pin 7

void setup() {
  Serial.begin(9600); // Start serial communication
}

void loop() {
  long duration, cm;
  // Send a short LOW pulse to ensure a clean HIGH pulse
  pinMode(pingPin, OUTPUT);
  digitalWrite(pingPin, LOW);
  delayMicroseconds(2);
  // Send a 5-microsecond HIGH pulse
  digitalWrite(pingPin, HIGH);
  delayMicroseconds(5);
  digitalWrite(pingPin, LOW);
  // Switch the pin to input mode and read the time
  pinMode(pingPin, INPUT);
  duration = pulseIn(pingPin, HIGH);
  // Convert time into distance
  cm = microsecondsToCentimeters(duration);

  // Print the distance in cm
  Serial.print("Distance: ");
  Serial.print(cm);
  Serial.println(" cm");
  delay(500); // Wait half a second before next reading
}

// Function to convert time to distance in cm
long microsecondsToCentimeters(long microseconds) {
  // Sound speed: 29 microseconds per cm (round trip)
  return microseconds / 29 / 2;
}
```

}

3. Find the Danger Zone:

- Test the sensor by placing objects at various distances.
- Decide the minimum safe distance (e.g., 20 cm) and set it as the trigger for a “danger” warning.

Possible Strategies for Implementation:

- **Model the Process First**
 - Teacher demonstrates how to connect the sensor and upload code before students start.
- **Pair Programming**
 - Students work in pairs to encourage collaboration—one connects the hardware while the other writes the code.
- **Incremental Testing**
 - Have students test the sensor connection first (with basic code) before adding the “danger zone” logic.
- **Use Visual Debugging**
 - Display the Serial Monitor on the board so students can compare readings with their own results.
- **Challenge Extension**
 - Ask students to modify the code so that an LED or buzzer activates when the danger zone is triggered.
- **Real-World Reflection**
 - Discuss how the same principle is used in parking sensors or robotics for obstacle avoidance.

Evaluate



Teacher's Role:

- Guide students as they upgrade their existing circuit with a vibration motor and expand their code to include Bluetooth or serial communication.
- Support students in understanding how the code works step-by-step, without giving direct answers.
- Ask open-ended questions like:
 - What does this line of code do?
 - How do you think the walker should respond when it senses something?
- Monitor safe and correct wiring of new components (e.g., vibration motor, Bluetooth).
- Encourage students to observe system behavior and suggest improvements.

- Normalize trial-and-error by praising effort and revision, not just final results. Provide visual or printed guides if needed (e.g., circuit diagrams, Bluetooth connection tips).
- Reconnect the activity to real-world purpose: Would this really help someone walking alone?

Expected from Students:

- Add a vibration motor to their Smart Walker circuit and test how it responds to obstacles.
- Fill in missing lines in the Arduino code and understand how the system works.
- Observe system behavior (e.g., does the motor vibrate when something is near?).
- Identify areas to improve like detection timing or vibration response.
- Reflect on their design: Would this system make someone feel safer?
- Document their work by sketching the circuit or explaining what each part does.
- Connect the idea of Bluetooth or alerts to real-world use (e.g., warning a caregiver).

Parts of the book included.



Showcase **Objective:**

Upgrade your Smart Walker by adding a vibration motor and editing your code so it can detect obstacles and send a warning to help keep users safe.

Possible Student Answers (Observe and Improve):

1. What happens when something moves in front of the sensor?

- The vibration motor turns on, and the Bluetooth sends a message.
- The walker gives immediate feedback when an obstacle is near.
- The sensor measures distance and triggers the vibration at a certain threshold.

2. What would you change or improve in your design?

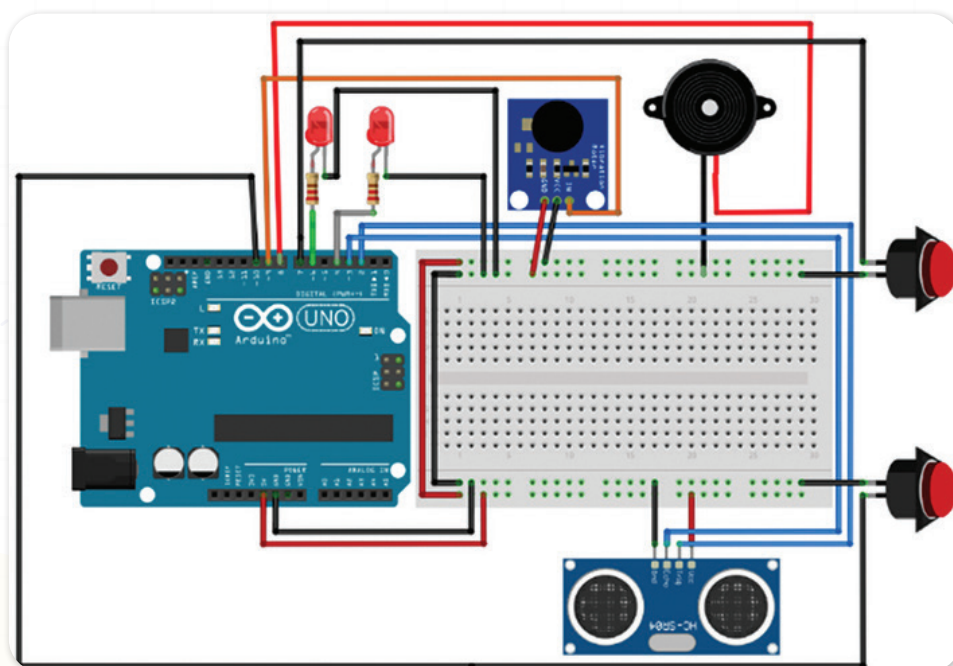
- Make the vibration stronger or last longer.
- Add sound or lights for additional feedback.

- Use a smaller sensor for a more compact design.
- Improve message clarity (e.g., “Obstacle ahead, please stop”).

Activity Instructions:

Circuit connection

Component	Pin on Arduino	Breadboard Connection Details
Ultrasonic Sensor HC-SR04	Trig → D10	Trigger pin (orange) to D10, Echo pin (yellow) to D11
	Echo → D11	
	VCC → 5V	Red wire to 5V power rail
	GND → GND	Black wire to ground rail
Red LEDs (x2)	D2, D3	Long leg → resistor → D2 and D3, Short leg → GND
Buzzer	D4	Positive → D4, Negative → GND
Vibration Motor (via Module)	D5	Signal pin (orange) → D5, VCC → 5V, GND → GND
Button 1 (left)	D6	One side to GND, other side to D6 (with INPUT_PULLUP)
Button 2 (right)	D7	One side to GND, other side to D7 (with INPUT_PULLUP)



Arduino Code:

```
const int trigPin = 10;
const int echoPin = 11;

const int led1 = 2;
const int led2 = 3;
const int buzzer = 4;
const int vibration = 5;

const int button1 = 6; // Panic Button
const int button2 = 7; // Headlight Toggle

bool headlightState = false;

void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);

  pinMode(led1, OUTPUT);
  pinMode(led2, OUTPUT);
  pinMode(buzzer, OUTPUT);
  pinMode(vibration, OUTPUT);

  pinMode(button1, INPUT_PULLUP);
  pinMode(button2, INPUT_PULLUP);

  Serial.begin(9600);
}

void loop() {
  // Distance Sensing
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
```



```
digitalWrite(trigPin, LOW);

long duration = pulseIn(echoPin, HIGH);
int distance = duration * 0.034 / 2;

Serial.print("Distance: ");
Serial.println(distance);

if (distance < 20) {
  digitalWrite(led1, HIGH);
  digitalWrite(buzzer, HIGH);
  digitalWrite(vibration, HIGH);
} else {
  digitalWrite(led1, LOW);
  digitalWrite(buzzer, LOW);
  digitalWrite(vibration, LOW);
}

// Panic Button pressed (active LOW)
if (digitalRead(button1) == LOW) {
  digitalWrite(buzzer, HIGH);
  digitalWrite(led2, HIGH);
  delay(1000);
  digitalWrite(buzzer, LOW);
  digitalWrite(led2, LOW);
}

// Headlight Button toggling
if (digitalRead(button2) == LOW) {
  headlightState = !headlightState;
  digitalWrite(led2, headlightState);
  delay(300); // Debounce
}
delay(200);
}
```

Possible Strategies for Implementation:

1. Pre-Activity Preparation

- **Teacher Role:**

- Review safety protocols for working with breadboards, wires, and vibration motors.
- Print or project the connection diagram and fill-in-the-code activity.
- Divide students into small teams (2–3 per group) for collaboration.

- **Strategies:**

- Use pre-labeled kits (each with Arduino, sensors, LEDs, buttons, motor module, etc.).
- Display a live demo or animation of how the system works before students begin.
- Allow students to explore sensors individually before combining them.

2. During Activity – Hands-On Building

- **Teacher Role:**

- Circulate between groups, check wiring and help troubleshoot.
- Prompt students with questions like:
“What do you expect the vibration to indicate?”
“Why do we need a delay between button presses?”

- **Strategies:**

- Encourage pair programming—one student types, one checks logic.
- Offer clue cards or QR code help sheets for students stuck on wiring/code.
- Let students swap tasks mid-activity (builder ↔ coder) for engagement.

3. Code Fill-in Strategy

- **Teacher Role:**

- Guide students in identifying missing code parts (variables, pin assignments, logic blocks).

- **Strategies:**

- Provide a partially filled template and let students guess or research missing parts.
- Offer a word bank of coding terms (e.g., pinMode, digitalRead, pulseIn).
- Create a mini-challenge: “Who can make the system alert only when objects are within 10 cm?”

4. Testing & Iteration

- **Teacher Role:**

- Encourage trial and error—students observe behavior and adjust values.
- Prompt reflection: “Does your buzzer react too early? Too late?”

- **Strategies:**

- Use a real-world scenario: "Imagine someone using this at night—what needs to change?"
- Run a "design pitch" where students test others' designs and give feedback.

Activity Rubric

Criteria	4–Exceeds Expectations	3–Meets Expectations	2–Approaching Expectations	1–Needs Improvement
Circuit Assembly	All components (ultrasonic sensor, vibration motor, LEDs, buzzer, buttons) are correctly connected and neatly arranged.	Most components are correctly connected with minimal errors.	Some components are connected but with multiple errors or unclear setup.	Circuit is incomplete or incorrectly wired; major help needed.
Code Accuracy	Code is complete, logical, and includes vibration + serial print + sensor reading + delays.	Code works with minor issues; includes basic functionality (sensor + feedback).	Code includes partial functionality but misses key logic or blocks.	Code is incomplete or non-functional; relies heavily on teacher assistance.
Testing & Debugging	Thorough testing done with multiple objects and conditions; student clearly refines the design.	Student tests and makes at least one meaningful change based on result.	Student tests once and makes minimal/no change.	Little or no testing observed.
Reflection (Observe & Improve)	Insightful responses about improvements and sensor behavior in real-life use.	Clear responses with some understanding of sensor behavior.	Basic responses with limited reflection.	Reflection incomplete or off-topic.
Teamwork & Collaboration	Actively collaborates, shares roles, and problem-solves respectfully.	Participates and communicates ideas within the team.	Participates occasionally; relies on others for help.	Rarely participates or disengaged.
Creativity & Innovation	Adds creative design features (e.g., custom alert signals, unique casing).	Design includes basic working features.	Design is functional but lacks personalization.	No creative input or design customization.

Elaborate



Teacher's Role:

- **Introduce the Task Briefly:** Explain the purpose of an emergency alert system and how such devices support elderly independence and safety.
- **Connect to Real-World Context:** Use a short scenario (e.g., “Imagine Grandma is alone and needs help quickly – how can tech help her call for assistance?”).
- **Model Possibilities (optional):** Show an example or brainstorm how a simple button + buzzer circuit works.
- **Clarify Expectations:** Go over deliverables (circuit plan, code sketch, explanation), emphasize safety at home, and offer optional materials support.

Expected from Students:

- Complete a functional prototype or simulation
- Submit a photo or screenshot of the circuit and code
- Provide a written explanation of their system's purpose and how it supports elderly safety
- Optional reflection on challenges and improvements

Parts of the book included.



Mastermind

Design a system where pressing a button triggers a buzzer alarm. This will help elderly users call for help in emergencies.

STEAM SCOUTS

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Mastermind

Objective:

Students will apply their understanding of input/output components and real-world problem-solving by designing a basic alert system. The system should activate a buzzer when a button is pressed, simulating an emergency help request for elderly individuals.

Activity Instructions

- **Circuit Prep:** Provide students with a diagram or allow them to take a photo of the actual classroom setup. Ensure the following connections are explained:

Component	Arduino Pin	Notes
Trig Pin	D2	Ultrasonic sensor trigger
Echo Pin	D3	Ultrasonic sensor echo
LED (Obstacle)	D4	Lights up if obstacle is detected
Vibration Motor	D5	Activates on obstacle detection
Headlight LED	D6	Toggle with button
Panic Button	D7	Triggers emergency mode
Buzzer	D8	Sounds during alert
Headlight Button	D9	Manual control for LED
Bluetooth RX	D10	SoftwareSerial TX
Bluetooth TX	D11	SoftwareSerial RX

- **Code Review:**
- Guide students to write the code themselves.
- Let students annotate/comment parts to show understanding.

Correct code:

```
// Pin Definitions
const int trigPin = 2;
const int echoPin = 3;
const int obstacleLED = 4;
const int vibrationPin = 5;
const int headlightLED = 6;
const int panicButton = 7;
const int buzzerPin = 8;
const int headlightButton = 9;

const int distanceThreshold = 40; // in cm

// State variables
bool panicActive = false;
bool lastPanicButtonState = HIGH;

bool headlightState = false;
bool lastHeadlightButtonState = HIGH;

// Define Bluetooth serial port (assuming you're using SoftwareSerial)
#include <SoftwareSerial.h>

SoftwareSerial bluetoothSerial(10, 11); // RX, TX (Connect Arduino TX to Bluetooth RX,
Arduino RX to Bluetooth TX)

void setup() {
  Serial.begin(9600);
  bluetoothSerial.begin(9600); // Initialize Bluetooth serial

  // Sensor and actuator pins
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(obstacleLED, OUTPUT);
  pinMode(vibrationPin, OUTPUT);
  pinMode(headlightLED, OUTPUT);
```

```

pinMode(buzzerPin, OUTPUT);

// Buttons with internal pull-up resistors
pinMode(panicButton, INPUT_PULLUP);
pinMode(headlightButton, INPUT_PULLUP);

// Initialize outputs LOW
digitalWrite(obstacleLED, LOW);
digitalWrite(headlightLED, LOW);
digitalWrite(vibrationPin, LOW);
digitalWrite(buzzerPin, LOW);

Serial.println("Smart Walker Initialized");
}

void loop() {
  // ==== Panic Button Toggle and Alert via Bluetooth ====
  bool currentPanicButtonState = digitalRead(panicButton);
  if (lastPanicButtonState == HIGH && currentPanicButtonState == LOW) {
    panicActive = !panicActive;
    Serial.println(panicActive ? "PANIC MODE ACTIVATED" : "Panic mode OFF");
    if (panicActive) {
      bluetoothSerial.println("E"); // Send "E" over Bluetooth when panic mode is activated
    }
    delay(300); // Debounce delay
  }
  lastPanicButtonState = currentPanicButtonState;

  // ==== Behavior Based on Mode ====
  if (panicActive) {
    emergencyAlert();
    // Potentially add a delay here or modify emergencyAlert to control its duration
  } else {
    checkObstacle();
  }
}

```

```

// ==== Manual Headlight Toggle ====
controlHeadlight();

delay(100);
}

void emergencyAlert() {
  Serial.println("Emergency Alert ON");
  tone(buzzerPin, 2000); // High-pitched continuous beep
  digitalWrite(obstacleLED, HIGH);
  delay(1000);
  noTone(buzzerPin);
  digitalWrite(obstacleLED, LOW);
  delay(300);
}

void checkObstacle() {
  int distance = getDistance();
  Serial.print("Distance: ");
  Serial.println(distance);

  if (distance > 0 && distance < distanceThreshold) {
    // Alert: Obstacle detected
    tone(buzzerPin, 1000); // Beep
    digitalWrite(obstacleLED, HIGH);
    digitalWrite(vibrationPin, HIGH);
    delay(200);
    noTone(buzzerPin);
    digitalWrite(obstacleLED, LOW);
    digitalWrite(vibrationPin, LOW);
    delay(200);
  } else {
    noTone(buzzerPin);
    digitalWrite(obstacleLED, LOW);
  }
}

```

```

    digitalWrite(vibrationPin, LOW);
  }
}

void controlHeadlight() {
  bool currentHeadlightButtonState = digitalRead(headlightButton);

  if (lastHeadlightButtonState == HIGH && currentHeadlightButtonState == LOW) {
    headlightState = !headlightState;
    digitalWrite(headlightLED, headlightState ? HIGH : LOW);
    Serial.println(headlightState ? "Headlight ON" : "Headlight OFF");
    delay(300); // Debounce
  }

  lastHeadlightButtonState = currentHeadlightButtonState;
}

int getDistance() {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

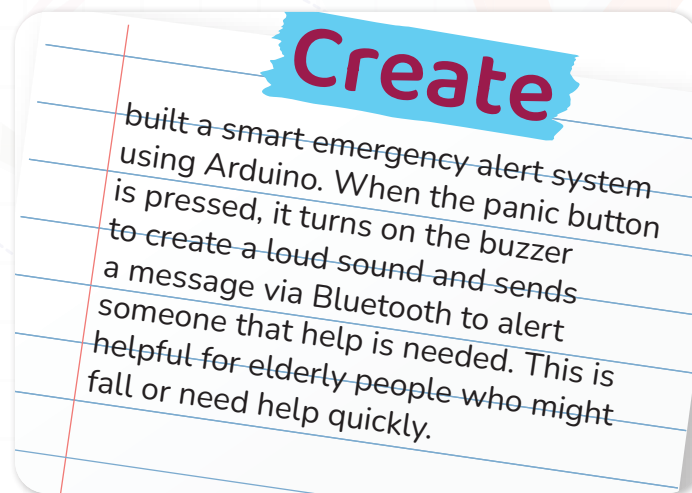
  long duration = pulseIn(echoPin, HIGH);
  int distance = duration * 0.034 / 2;
  return distance;
}

```

- **Assignment Requirements:**

- Upload photo or screenshot of the circuit
- Submit the Arduino code file

Don't forget to go back to the EDP page and add these to create part.



Now I Can

Objective:

Students will reflect on the knowledge and skills they developed during the unit by identifying what they are now able to do as a result of hands-on learning, coding, and circuit building. This builds self-awareness, reinforces learning outcomes, and prepares them to communicate their achievements.

Activity Instructions:

For Teachers:

1. Review and Recap

- Guide a class discussion reviewing each bullet point.
- Ask students to give real examples from their Smart Walker project (e.g., “Which sensor worked best in your test?” or “How did your vibration motor help the user?”).
- Optionally, ask them to write a short explanation or draw a diagram for one of the bullet points.

2. Optional Pair Share or Gallery Walk:

- Let students turn and talk with a partner to discuss which skill they are most proud of and why.
- You may also display their statements on a reflection board or poster titled “Look What We Can Do!”

Relation to Standards

1. ISTE Standards for Students (International Society for Technology in Education)

- **1.1 Empowered Learner:**
 - Students use technology to take an active role in achieving and demonstrating competency in their learning goals.
 - Students build, test, and choose sensors based on data from their prototype testing.

- **1.4 Innovative Designer:**

- Students use a variety of technologies within a design process to identify and solve problems by creating new solutions.
- Designing the Smart Walker circuit and programming features like alert systems for elderly users demonstrates innovative thinking.

- **1.5 Computational Thinker:**

- Students develop and employ strategies for understanding and solving problems using technology.
- They apply programming logic to sensors, inputs, and Bluetooth communication to solve real-world challenges.

- **1.7 Global Collaborator:**

- Students use digital tools to broaden their perspectives and enrich their learning by collaborating with others and working effectively in teams.
- Students reflect on user needs (elderly users) and discuss possible improvements and communication strategies.

2. NGSS – Next Generation Science Standards (Engineering Practices)

- **3-5-ETS1-1:**

- Define a simple design problem reflecting a need or a want that includes specified criteria for success and constraints on materials, time, or cost.
- Students identify elderly mobility challenges and design an assistive technology prototype.

- **3-5-ETS1-2:**

- Generate and compare multiple possible solutions to a problem based on how well each is likely to meet the criteria and constraints of the problem.
- By testing ultrasonic and IR sensors, students evaluate effectiveness in different conditions.

3. CSTA K–12 Computer Science Standards

- **3B-AP-21:**

- Develop and use a series of test cases to verify that a program performs according to its design specifications.
- Students observe and improve their Smart Walker code after testing it in multiple scenarios.

- **3B-CS-02:**

- Illustrate how sensors, devices, and networks collect and transmit data.
- Students build circuits with ultrasonic sensors and Bluetooth modules that detect obstacles and transmit messages.

4. UN SDGs – Sustainable Development Goals

- **Goal 3:** Good Health and Well-being
 - Designing assistive technology for elderly people helps improve quality of life, independence, and safety.
- **Goal 9:** Industry, Innovation, and Infrastructure
 - Students engage in hands-on innovation using engineering and electronics to develop impactful solutions.